

Instructions for Project 2: Estimation from a Simple Random Sample

In this project, you will use an interval to estimate a population proportion or mean using data collected from a random probability sampling method. Please complete the following steps, then write a one page summary of your results following the grading rubric.

Important Dates:

Thursday, March 12: R code and simple random sampling are introduced in lab

Thursday, March 26: Peer review of sampling method and interval choice in lab

Sunday, March 29th: one page summary and R notebook due in Canvas

Steps:

1. Think of a population you could list out, randomly sample from the list, and obtain data on a single variable (either a number or a binary outcome) from the items/people in the random sample. It's hard to randomly sample people because permission to conduct research on people must be obtained from the university and because it's hard to get access to a list of contact information for all people in a given population. For these reasons, I recommend using a population of non-human objects to sample from. For example, you could "list" all parking spaces in the parking lot by looking at an image from Google Maps, numbering the spaces and then randomly selecting 10 spaces from the list to visit and collect one piece of information at (that is, collect a single variable, e.g. is the space occupied or not? Is the auto a truck or not? Is the auto white or not? What year is the auto from?). You could also sample from a list of items on a website, like I did for the sample project.

My population is _____, which has $N =$ _____ objects.

2. Once you've decided on a population, think of what mean or proportion you'd like to estimate for this population. For example, you might want to estimate the proportion of cars at CPH that are white, or the mean year of autos at CPH. What is the variable that corresponds to the proportion or mean? This variable is the information that you will need to collect for each object in your sample. For example, if you're interested in estimating the mean year, you'll need to collect a single numerical variable, "year" from each auto in your sample.

My variable is _____, which is _____ binary categorical _____ numerical (circle one) _____
so I'll be estimating a _____ proportion _____ mean (circle one) _____.

3. Now decide on a sample size. Generally, the more objects in your sample, the more precision you'll have in your estimate. But collecting data from more objects takes more time so we must balance time and precision. Professionals who collect data perform a "sample size calculation" to determine the optimal balance between these two factors. In this project, I'll just ask you to consider collect info from 10 or the largest number of objects you could visit in 1-2 hours of time and use that as your sample size.

I think the largest number of objects I could feasibly visit and collect information for my variable is $n =$ _____

4. Now use a random number generator to randomly select n items from the list of N objects in your population. I've used the one at RossmanChance.com/applets in class:

<https://www.rossmanchance.com/applets/2021/randomnumbers/GenRandom01.htm>

The n items I need to visit are:

5. Now visit each of the n items in your sample and collect your variable. The dataset from my sample is:

6. Now compute the mean or proportion from your dataset. This is your single best estimate for the population mean or proportion. My single best estimate of the population is _____

7. Now decide which of the five situations from Lecture 19 that your data is in. If you have a finite value of N , then you'll want to multiply the SE by the FPC factor. If you have a mean, you'll want to use the SE from the t-interval. If you have a proportion, you'll want to use the SE from the one proportion z-interval (Wald) if the sample size is large enough or the binomial if the sample size is not large. Record and check the conditions for your selected interval carefully, then use the R code I provided in Demo for Lab 7 or Project 2 to compute it using your data.

My 95% confidence interval estimate of the population mean or proportion is $L =$ _____ and $U =$ _____.

The conditions for this interval are _____

8. Now write an interpretation of your confidence interval estimate in context using the following "recipe":

I am 95% confident that the _____ (parameter, e.g. mean or proportion) of
_____ (variable) for _____ (population) is between _____ (L) and _____ (U).

Sample Report for Project 2: Estimation from a Random Sample

Introduction:

What is the mean age of high schools in California? As someone who teaches college level classes, I was interested in finding out more about the population of schools my students might be coming from. One question is how old their schools tend to be: I think older schools might tend to have more resources and stronger connections with a community, whereas newer schools might have less resources.

Methods

Data Collection:

On April 3rd, 2024, I visited www.cde.ca.gov/SchoolDirectory and searched for any type of high school. I found N=1617 schools. I used the random number generator on RossmanChance.com to choose n=10 numbers from 1 to 1617. I located each of the corresponding schools by setting the number of schools shown to be 100, going to the page with the right hundreds value and counting up or down to find the school corresponding to each random number's 10th and 1th place. I then looked up the "Open Date" for the corresponding 10 schools. I calculated the age of each of these schools in 2024 by subtracting the year of the "Open Date" from 2024.

Data Analysis

I used the survey library in R to compute the 95% confidence t-interval with finite population correction for the mean age of all high schools in CA using the ages from my dataset of 10 schools. The conditions for using this interval are having a simple random sample and believing the population of ages is roughly symmetric.

Results

Descriptive Statistics

The mean age of the ten schools was 35 years and the standard deviation was 25 years. A histogram of the ages is shown in Figure 1:

Inferential Statistics

Based on my dataset, I'm 95% confident that the mean age of all CA high schools is between 20 and 51 years old in 2024.

Conclusion

The mean age of high schools in CA was estimated to be younger than I was expecting. Because my dataset of 10 schools was chosen according to a simple random sampling scheme, I don't expect that my estimate is biased. However, my sample size of 10 is pretty small and so my estimate is not very precise: there's about a 35 year difference between the lowest and highest reasonable estimates for the mean age. I could improve the precision by increasing the sample size.

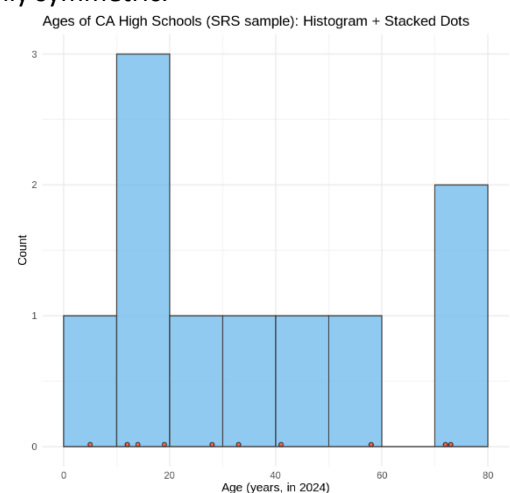


Figure1 : Ages in 2024 of 10 randomly selected CA high schools.

Grading Rubric for Project 2

Component	Description	Points
1. Introduction	- Briefly describe in a single sentence the parameter (mean or proportion) you're estimating. - Clearly state the population associated with this parameter. - Briefly describe your motivation for choosing to estimate this quantity.	3
2. Methods		
a. Data Collection	- Include a concise description of your population, including the size. - Describe how you obtained a simple random sample from a list of the population (e.g. used a random number generator and the population size). - Be sure to describe exactly how you located an item corresponding to one of your random numbers (for example, I set the schools shown on a page to be 100...) - Describe the variable (information) you collected from each item in your sample: be clear as to whether you collected a number or a binary outcome. If a number be sure to say it's units.	8
b. Data Analysis	- Describe how you computed the 95% confidence interval for the parameter you're estimating in R (e.g. used survey or t.test or prop.test or binom.test) - State whether or not you used the FPC - Describe any conditions for your selected interval to be valid.	6
3. Results		
a. Descriptive Statistics	- Include a well-labelled graph of your dataset - Include the mean and SD if numerical variable, or count and proportion if binary categorical.	6
b. Inferential Statistics	- Include the values of L and U from your 95% confidence interval estimate - Include a one sentence interpretation of the interval estimate in the context of the parameter and population you have	6
4. Conclusion	- Briefly describe your reaction to the final estimate - is the estimate about what you expected? - Briefly state whether or not you believe that your estimate is biased - Briefly state how you could make your estimate more precise	6